EXCERPT Sections 6.3 – 6.4 of the Sol Protocol Game Design Document

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6 **Ship Systems**

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6.5 System Chips & Mods, not included in this excerpt

… continued from 6.2 Energy Management

6.3 Repairs

Status	Implemented (Alpha) · v2
Discipline	Systems Design · Level Design
Related	6.4 Ship Generator & Power Cells · 7.2 Loot (Repair Spheres) · 8.1 Playable Tutorial

6.3.1 Overview

Inside a dedicated room on the ship there are multiple control panels, each one representing a specific system or weapon. When a system is damaged it is repaired from its panel, using repair spheres collected as loot during exploration.

6.3.2 Panel Layout

- Each panel represents an entire system: its name, its installed power-ups (mods) and its current status.
- Each panel includes the **end of a tube**, sealed with a **handle-like counter**.
- Panels are visually distinct and labelled according to the system or weapon they represent.

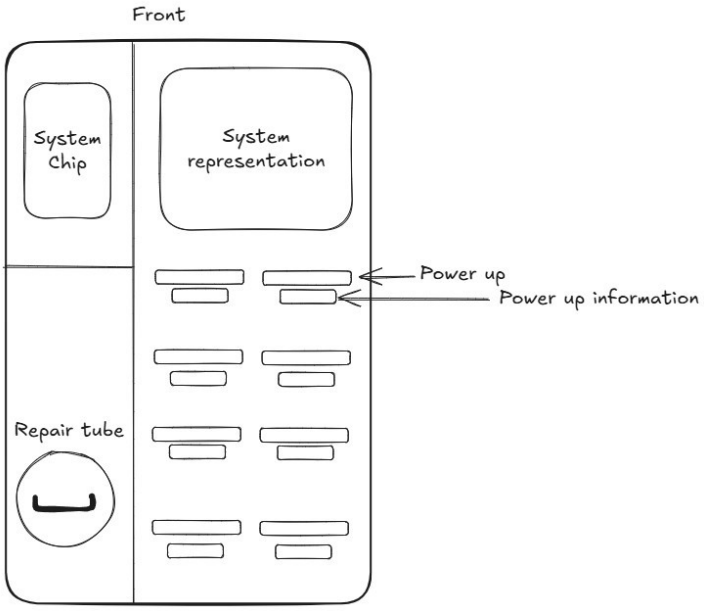


Figure 6.11 — System panel layout concept (front view): system chip slot, system representation screen, power-up slots with their info labels, and the repair tube.

6.3.3 Damage Feedback

When a system is damaged:

- an **alert sound** is triggered;
- a **red warning light** illuminates the corresponding system panel.

6.3.4 Repair Process

To start a repair, the player must:

1. **Open the tube** upwards by interacting with the handle counter.
2. **Insert a repair sphere** (containing nano-bots) into the tube.
3. **Close the handle counter** to seal the tube.

Once the tube is closed, the nano-bots are released into the system through the tube and begin the **automatic repair** of the damaged component.

6.3.5 Repair Sphere Types

- There are multiple types of repair spheres, each with **different healing values**.
- The effectiveness of a repair depends on the **type of sphere** used.
- **Hull repair sphere** (common–rare): repairs part of the damage sustained by the ship hull.
- **System repair sphere** (common, blue): repairs all the damage sustained by a single system or weapon.
- *Optional design extension:* some spheres might be rarer, more expensive, or only applicable to certain system types.

6.3.6 Revision History

Version	Milestone	Repair mechanic	Status
v1	First Playable	Cable panel per system: any damage breaks the cables. The player puts a hand inside the panel and holds grip for 5 s; cables snap back and the system returns to 100% health.	Deprecated
v2	Alpha	System panels in a dedicated room, repaired with repair spheres (this section).	Current

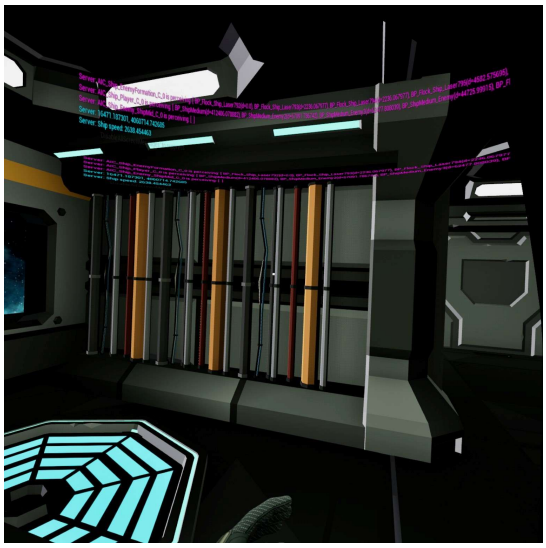


Figure 6.12 — v1 (deprecated) — cable panel, system at 100% health.

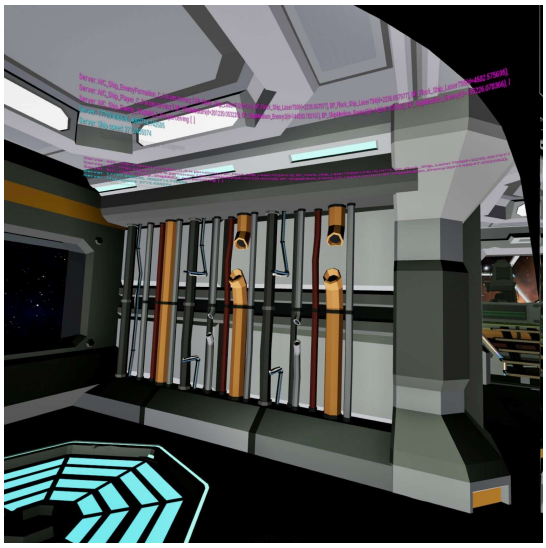


Figure 6.13 — v1 (deprecated) — broken cables, system below 100%.

6.3.7 Current Implementation (Alpha Build)

Element	Concept	Alpha build
System chip	Slot on the top-left	Top-left, colour-coded by system type
System representation	Screen with system status	Screen + health bar
Power-up (mod) slots	8 slots with info labels	4 slots
Repair tube	Open → insert sphere → close	Round socket: the sphere is placed in it and absorbed automatically



Figure 6.14 — *Systems room, Alpha build — one panel per ship system and weapon.*



Figure 6.15 — *System panel, Alpha build: system chip, screen and health bar, mod slots, repair socket.*

Open Tasks

- Rename repair spheres to a nanobot-themed resource (lore consistency).
- Building or re-installing a system consumes repair resources (removes the free-repair exploit).

6.4 Ship Generator & Power Cells

Status	Generator implemented (Alpha) · power-cell progression in design
Discipline	Systems Design · Level Design
Related	6.2 Energy Management · 6.3 Repairs · 7.2 Loot

6.4.1 Overview

The ship generator is located at the rear of the medium-sized ship. It features 15 slots where power cells can be inserted to increase the ship's energy capacity. Each power cell fits into a specific socket, and each socket can hold only one cell at a time.

Not all sockets are available from the start of the game, so unusable sockets are covered by a cap. As the player progresses, new sockets are unlocked and the ship becomes more powerful.

The screens on the generator display the number of installed cells and the maximum energy capacity that can be reached.

6.4.2 Design Intent

Energy capacity is a progression axis. Runs start with low energy; power cells are a loot type, and every cell the crew finds and installs raises the ship's maximum energy.

- Power-cell insertion mechanic.
- Adjustable power output.

6.4.3 Power Cells

The power cells can mix the battery look of Subnautica with the loot crates shown in the first trailer (no longer used in game). They should have a distinctive, unusual shape that gives players immediate visual feedback about where they must be inserted.



Figure 6.16 — *Reference — sci-fi power cell.*



Figure 6.17 — *Reference — SM_Prop_Barrel_02, loot crate from the first trailer.*

Rule
Whatever design is chosen, the slots must NOT be simple circles or squares. They need a distinctive shape so that they are memorable and instantly recognisable to the player.

6.4.4 Placement

The generator is placed near the empty room (the future elevator area), directly in front of the entrance to the med bay.

- The generator must be deep enough for a power cell to be inserted; cells snap in like the system chips in the system panels.
- The room is also the access point for the future elevator, so the console cannot take up too much space or block the passage.

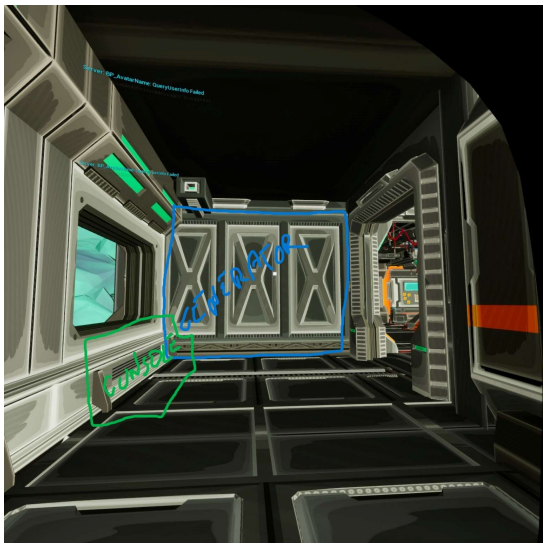


Figure 6.18 — Corridor view with generator (blue) and console (green) blocked out.

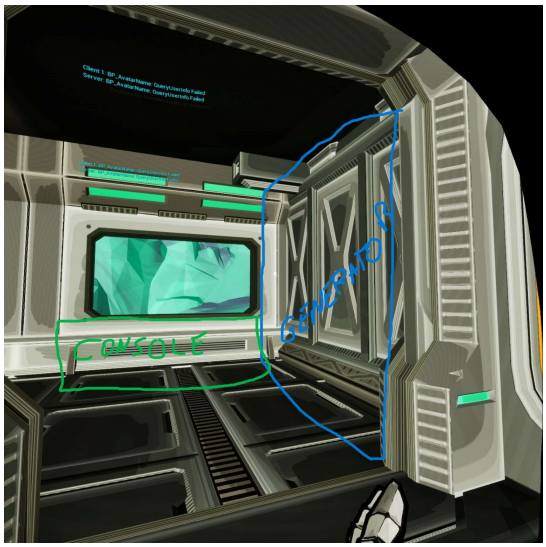


Figure 6.19 — Close-up of the chosen wall for generator and console.

6.4.5 Generator Layout Options

> **Screens**
> The screens must show text and data, so they stay simple and clean: flat colours, no extra visual elements or complex details (same approach as the system panels).

Option A — Slots around a central screen

Similar to the reference, but with many more power-cell slots arranged around a central screen.



Figure 6.20 — Reference — Subnautica power-cell charger.

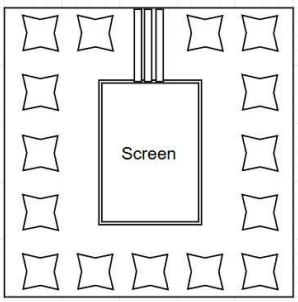


Figure 6.21 — Option A layout sketch.

Option B — Wall-integrated grid

A wall-integrated generator with a large screen on top and a grid of slots below (fewer slots than the reference).

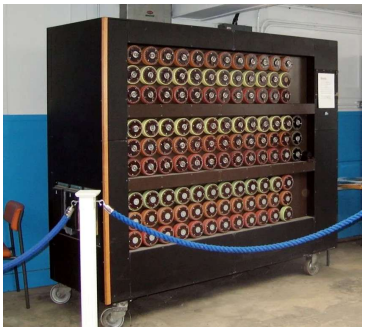


Figure 6.22 — Reference — Bombe machine, Bletchley Park.

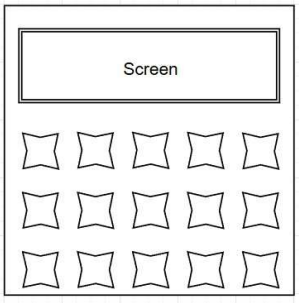


Figure 6.23 — Option B layout sketch.

6.4.6 Adjacent Console

- **Current scope:** environment piece with two screens.
- **Future scope:** levers and buttons around the screens to host repair minigames during combat, raising tension and player engagement.



Figure 6.24 — Reference — mission control console.

6.4.7 Current Implementation (Alpha Build)

Element	Concept	Alpha build
Layout	Option A or Option B	Option B, wall-integrated; screen on the side panel
Sockets	15, distinctive shape	15 in three columns of five, angular shape
Locked sockets	Covered by a cap	Hazard-striped caps
Screen	Installed cells and max capacity, flat colours	Side screen, flat colour

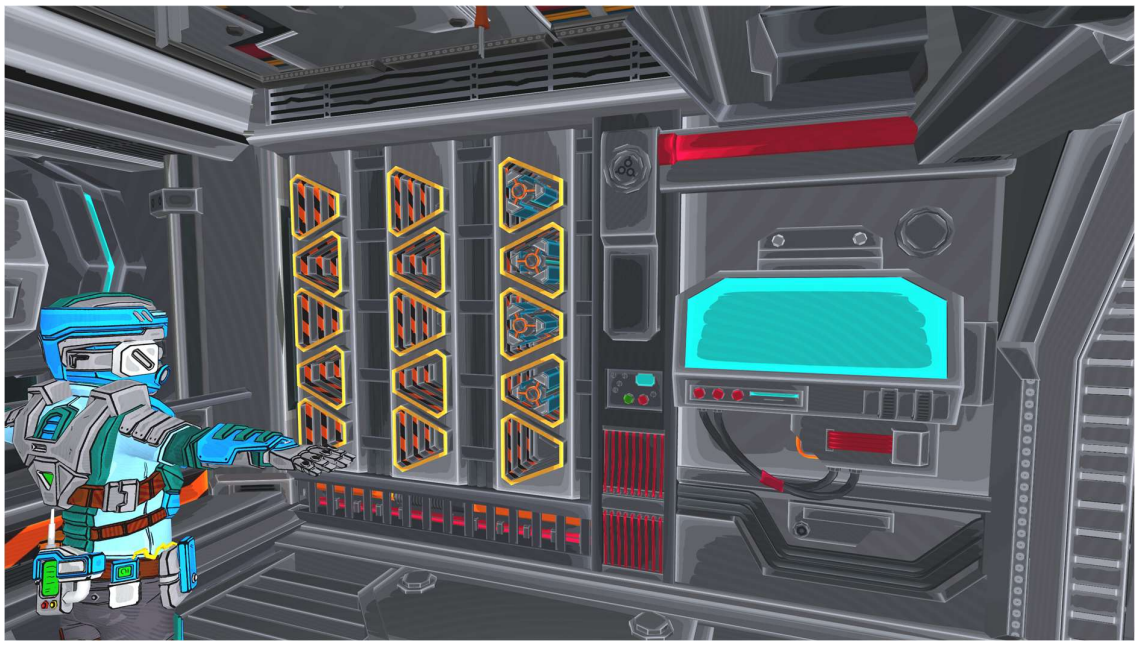


Figure 6.25 — *Ship generator, Alpha build.*

Continues in 6.5 System Chips & Mods →